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SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS

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Review III

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TITLE: Security Intelligence and Public Sentiment Analysis on Social Media Platforms

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DECLARATION

We hereby declare that the report entitled "**FORENSIC ANALYSIS ON SOCIAL MEDIA**" submitted by us, for the **CSE3501** and **Information Security Analysis and Audit** to Vellore Institute of Technology is a record of bonafide work carried out by me under the supervision of **ANBARASA KUMAR A**.

We further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for any other courses in this institute or any other institute or university.

Place : Vellore

Date : 06/11/2025

Signature of the Candidate 1:

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Signature of the Candidate 2:

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Abstract:

Social media platforms are now a major part of everyday life, helping people connect, share information, and express themselves. However, they also contain a huge amount of data that can be very useful for legal, investigative, and security purposes.

This project focuses on using Open-Source Intelligence (OSINT) tools to perform a detailed digital forensic investigation on social media, specifically targeting Instagram as the platform for analysis. The main goal is to use the OSINT framework-a collection of free tools and resources to gather user data from Instagram. This framework also simplifies the process of collecting relevant information, making investigations more efficient and effective.This project mainly focuses on extracting security related insights and analyzing public emotions expressed on social media using Python-based Machine Learning techniques.

KeyWords:

Social Media, Digital Forensics, OSINT, Instagram, Data Collection, Investigation, Information Extraction, Open-Source Intelligence, Machine Learning Techniques.

Introduction:

In recent years, rapid technological growth has transformed how people communicate and do business, with Instagram playing a key role in this digital transformation. The platform has not only made personal interactions easier but has also provided new and creative ways for businesses to promote their products and services to a global audience. As a result, Instagram has evolved from being just a space for personal sharing into a powerful tool for advertising, sales, and brand promotion.However, this growing dependence on Instagram has also brought an increase in potential risks that, if exploited, could cause serious financial and reputational harm to individuals and organizations. Alongside this digital revolution, advancements in technology have led to larger storage capacities and massive data generation, making digital forensic investigations more complex and time-consuming. Therefore, understanding where and how Instagram stores its digital traces has become essential for speeding up forensic investigations and supporting the quick resolution of criminal cases.

Social media serves as a massive repository of user generated data where opinions, news, and reactions to events are shared instantly. These online discussions often reflect the public's mindset during crises, social movements, or security-related situations. However, manually analyzing this high-volume and unstructured data is nearly impossible.
To address this, the proposed system uses Python-based data processing and Machine Learning sentiment analysis algorithms to examine emotional tones and identify potential

security concerns. The project integrates natural language understanding with predictive analytics to transform raw text into structured intelligence. This contributes to improved decision-making and situational awareness for authorities and organizations monitoring online environments.

Objective:

The main objective of this project is to analyze and understand how Instagram manages, stores, and shares user data, using Open-Source Intelligence (OSINT) tools for digital forensic purposes. It aims to identify and extract digital traces that can assist in investigations, legal proceedings, and security assessments.

The primary objective of this project is to design and implement a Python-based system capable of performing security intelligence and sentiment analysis on social media platforms. The system aims to automatically gather large volumes of public data such as posts, comments, and tweets, and process them using Machine Learning techniques. Through text preprocessing, feature extraction, and classification algorithms, the project seeks to identify the emotional polarity of user opinions as positive, negative, or neutral. Additionally, it focuses on detecting posts containing security-related keywords or potential threat indicators to assist in early awareness and response. By visualizing sentiment and security trends through graphical dashboards, the project contributes to enhanced understanding of public behavior and supports decision making processes for organizations and authorities monitoring online environments.

Problem statement:

With the rapid advancement of digital technology, Instagram has become a leading platform for communication, business promotion, and personal expression. However, its widespread use also results in a vast amount of data that can potentially be exploited for malicious or illegal activities. Conducting forensic analysis on this data is challenging because of its complex structure, massive scale, and constantly changing nature.

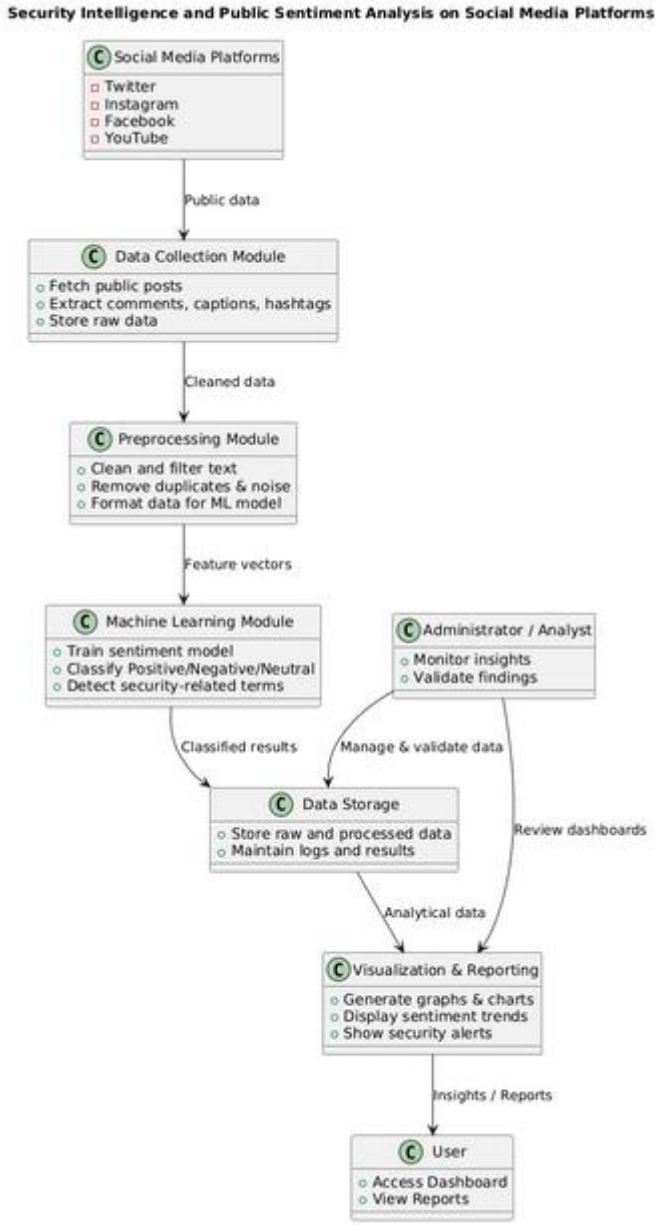
The core issue is the absence of effective techniques and tools to accurately locate, collect, and analyze digital evidence from Instagram. This limitation makes it difficult for investigators and law enforcement professionals to obtain trustworthy information quickly. Hence, there is a strong need for a comprehensive and efficient framework that utilizes Open-Source Intelligence (OSINT) tools to streamline and enhance the process of digital forensic investigation on Instagram. Therefore, there is a clear need for an automated Python-based system that uses Machine Learning algorithms to perform sentiment analysis and detect posts linked to possible security concerns. This approach can help organizations and authorities gain better insight into public behavior, identify potential risks, and take preventive measures when necessary.

Literature Review:

S.No	Title	Author Details	Methodologies Used	Merits	Demerits
1.	Investigating methods for forensic analysis of social media data to support criminal investigations	M Arshad et al., 2025	AI, ML, mixed-methods, NLP (BERT), network analysis	Scalable, accurate, case-based	High data/computation needs
2.	A hybrid approach for social media forensics	Ashfaq Ahmad , 2025	ML algorithms to detect offensive language, sentiment classification	Detects cyber threats, automated	May miss hidden context
3.	Machine Learning Techniques for Social Media Investigation	AAS Mahmoud, 2025	Hybrid digital forensic models, universal device compatibility	Device-wide applicability	May lack platform-specific detail
4.	Research Papers in Digital Forensics 2025	Nishchal Soni, 2025	Survey of trending forensic tools and techniques	Broad overview, latest trends	Not all studies peer-reviewed
5.	Digital Forensics: Confronting Modern Cyber Crimes	Herald Open Access, 2025	Advanced algorithms, timeline and behavioral analysis	Multi-crime focus	Rapid tech changes
6.	Article: A hybrid approach for social	AAS Mahmoud, 2025	Hybrid forensics for web devices	Cross-platform	Gaps in deep analysis

	media forensics				
7.	Digital forensics for the socio-cyber world (DF-SCW): A novel framework for deepfake multimedia investigation on social media platforms	Basumatary , , 2025	Deepfake detection, novel framework	Multimedia forensics	May not cover text/normative issues
8.	Digital Forensics and Media Offences – Investigate Synergy in the Current Landscape	Kalita , 2025	Data recovery, deleted communications, timeline analysis	Recovers deleted social data	Access/legal concerns
9.	Forensic Investigations of Social Media Apps: Tools and Techniques	NBFtools, 2025	Tool reviews, legal challenges, metadata analysis	Real-world tool focus	Tool access limits
10.	Mobile Device Forensics For Social Media Investigations	SSRN, 2024	Mobile device extraction, algorithmic analysis	Mobile-centric, efficient	Device/OS compatibility

Architecture:



Proposed Methodologies:

The proposed methodology adopts a structured and ethical process for conducting forensic analysis on Instagram using OSINT tools. The investigation begins by defining its scope and identifying specific public profiles or posts to be examined. Authorized and policy-compliant extraction tools are then employed to gather relevant information such as captions, comments, and user details.

Collected data is systematically organized into categories like posts, user information, and timestamps to establish activity patterns and possible event timelines. The analysis focuses on identifying suspicious keywords, unusual communication patterns, or irregular user behaviors that could indicate potential security concerns. To ensure reliability, extracted information is cross-checked with external sources and validated through multiple verification tools.Each stage of the process from data acquisition to analysis—is carefully documented to maintain transparency and evidential integrity. The final report summarizes major findings, highlights potential risks, and presents results through clear visual representations such as graphs or timelines. This methodology ensures that the forensic analysis remains accurate, compliant, and effective in supporting digital investigations and continuous monitoring of online activities.

Coding and Implementation:

This chapter presents the detailed coding and implementation process of the project titled "Information Security Audit and Sentiment Analysis of Instagram Data Using Python". The implementation was carried out using Python within the Windows Subsystem for Linux (WSL) environment. The project aims to analyze public sentiment from Instagram comments to support information security auditing and digital intelligence assessment.

Tools and Technologies Used

- Programming Language: Python 3
- Environment: Windows Subsystem for Linux (Ubuntu)
- Libraries: Instaloader, Pandas, TextBlob
- IDE/Editor: Visual Studio Code / Nano Editor
- Data Source: Instagram Public Profiles
- Output Format: CSV report (sentiment_report.csv)

Implementation Workflow

The implementation of the project was carried out in several key steps as described below:

Step 1: Environment Setup

Python and required libraries were installed within WSL. A virtual environment was created to manage dependencies.

Step 2: Data Collection

Instagram data was collected using the Instaloader library. Public profiles were accessed and metadata, captions, and comments were downloaded. A temporary delay and rate-limiting

mechanism were handled to comply with the platform's restrictions.

Step 3: Data Preprocessing
The downloaded text data (comments/captions) were cleaned and stored in text files for analysis.

Step 4: Sentiment Analysis
A custom Python script (predict.py) was implemented using the TextBlob library to perform polarity-based sentiment analysis. Each comment was categorized as Positive, Negative, or Neutral based on its polarity score.

Step 5: Result Generation
Results were displayed in tabular format on the terminal and exported as a structured CSV report named 'sentiment_report.csv'. This report supports interpretation and visualization of social media sentiment for auditing purposes.

Source Code
The following code segment represents the main sentiment analysis logic implemented in the project:

```
import pandas as pd
from textblob import TextBlob
import os

def analyze_sentiment(text):
    blob = TextBlob(text)
    polarity = blob.sentiment.polarity
    if polarity > 0.1:
        return 'Positive', round(polarity, 3)
    elif polarity < -0.1:
        return 'Negative', round(polarity, 3)
    else:
        return 'Neutral', round(polarity, 3)

def analyze_file(filepath):
    with open(filepath, 'r', encoding='utf-8') as f:
        lines = [line.strip() for line in f if line.strip()]
    results = []
    for i, comment in enumerate(lines, start=1):
        label, polarity = analyze_sentiment(comment)
        results.append({'ID': i, 'Comment': comment, 'Sentiment': label, 'Polarity': polarity})
    return results

def main():
    folder = input('Enter folder path or leave blank for current directory: ').strip() or '.'
    all_results = []
    for file in os.listdir(folder):
        if file.endswith('.txt'):
            path = os.path.join(folder, file)
            print(f'Analyzing {file} ...')
            all_results.extend(analyze_file(path))
```

```
if not all_results:
        print('No text files found! Please make sure you have .txt files with comments.')
        return

    df = pd.DataFrame(all_results)
    print(df)
    df.to_csv('sentiment_report.csv', index=False)
    print('\nSentiment analysis complete! Results saved to sentiment_report.csv')

if __name__ == '__main__':
    main()
```

Output Description

The system outputs both a terminal-based table and a CSV file named 'sentiment_report.csv'. Each row in the report contains the comment ID, the original text, the detected sentiment, and the polarity score. An example of the output is shown below:

ID | Comment | Sentiment | Polarity
1 | I love this product! | Positive | 0.625
2 | This post is terrible. | Negative | -1.000
3 | It's okay, not bad. | Neutral | 0.000

Phy code:

```
  GNU nano 7.2                                                predict.py
import pandas as pd
from textblob import TextBlob
import os

def analyze_sentiment(text):
    """Return sentiment label and polarity value."""
    blob = TextBlob(text)
    polarity = blob.sentiment.polarity
    if polarity > 0.1:
        return "Positive", round(polarity, 3)
    elif polarity < -0.1:
        return "Negative", round(polarity, 3)
    else:
        return "Neutral", round(polarity, 3)

def analyze_file(filepath):
    with open(filepath, "r", encoding="utf-8") as f:
        lines = [line.strip() for line in f if line.strip()]
    results = []
    for i, comment in enumerate(lines, start=1):
        label, polarity = analyze_sentiment(comment)
        results.append({"ID": i, "Comment": comment, "Sentiment": label, "Polarity": polarity})
    return results

def main():
    folder = input("Enter folder path or leave blank for current directory: ").strip() or "."
    all_results = []

    for file in os.listdir(folder):

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

USER CREDENTIALS:

```
yuktha@yukthasri: ~/Projects
  GNU nano 7.2                    config/credentials.ini *
[Credentials]
username =yukthasri11
password =Yukupuni@2005
hikerapi_token =

^G Help      ^O Write Out  ^W Where Is   ^K Cut      ^T Execute   ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste    ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy
```

INPUT COMMENTS:

```
yuktha@yukthasri: ~/Projects
  GNU nano 7.2                    comments.txt
I love this photo!
This post is boring.
It’s okay, not great.
Such a beautiful picture.
I'm not impressed.

                          [ Read 5 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify
```

POLARITY CHECK:

```
docker_reqs.txt     visualize_results.py
main.py             visualize_results.py.save
output              yukt.harochna123
yuktha@yukthasri:~/Projects/Osintgram$ source venv/bin/activate
(venv) yuktha@yukthasri:~/Projects/Osintgram$ python3 predict.py
python3 visualize_results.py
Enter folder path or leave blank for current directory:
Analyzing requirements.txt ...
Analyzing docker_reqs.txt ...
Analyzing comments.txt ...
    ID  ... Polarity
0    1  ...    0.000
1    2  ...    0.000
2    3  ...    0.000
3    4  ...    0.000
4    5  ...    0.000
5    6  ...    0.000
6    7  ...    0.000
7    1  ...    0.000
8    2  ...    0.000
9    3  ...    0.000
10   4  ...    0.000
11   5  ...    0.000
12   6  ...    0.000
13   1  ...    0.625
14   2  ...   -1.000
15   3  ...    0.050
16   4  ...    0.425
17   5  ...   -0.500

[18 rows x 4 columns]

✅ Sentiment analysis complete! Results saved to sentiment_report.csv
```

SENTIMENT ANALYSIS:

```
(venv) yuktha@yukthasri:~/Projects/Osintgram$ python3 predict.py
Enter folder path or leave blank for current directory:
Analyzing requirements.txt ...
Analyzing docker_reqs.txt ...
Analyzing comments.txt ...
    ID                                        Comment Sentiment  Polarity
0    1                     requests-toolbelt==0.9.1   Neutral     0.000
1    2                                 geopy>=2.0.0   Neutral     0.000
2    3                            prettytable==0.7.2   Neutral     0.000
3    4                  instagram-private-api==1.6.0   Neutral     0.000
4    5  gnureadline>=8.0.0; platform_system != "Windows"   Neutral     0.000
5    6     pyreadline==2.1; platform_system == "Windows"   Neutral     0.000
6    7                              hikerapi==1.7.1   Neutral     0.000
7    1                     requests-toolbelt==0.9.1   Neutral     0.000
8    2                                 geopy>=2.0.0   Neutral     0.000
9    3                            prettytable==0.7.2   Neutral     0.000
10   4                  instagram-private-api==1.6.0   Neutral     0.000
11   5                            gnureadline>=8.0.0   Neutral     0.000
12   6                              hikerapi==1.7.1   Neutral     0.000
13   1                            I love this photo!  Positive     0.625
14   2                         This post is boring.  Negative    -1.000
15   3                        It's okay, not great.   Neutral     0.050
16   4                     Such a beautiful picture.  Positive     0.425
17   5                            I'm not impressed.  Negative    -0.500

✅ Sentiment analysis complete! Results saved to sentiment_report.csv
(venv) yuktha@yukthasri:~/Projects/Osintgram$
```

```
✅ Sentiment analysis complete! Results saved to sentiment_report.csv
(venv) yuktha@yukthasri:~/Projects/Osintgram$ cat sentiment_report.csv
ID,Comment,Sentiment,Polarity
1,requests-toolbelt==0.9.1,Neutral,0.0
2,geopy>=2.0.0,Neutral,0.0
3,prettytable==0.7.2,Neutral,0.0
4,instagram-private-api==1.6.0,Neutral,0.0
5,"gnureadline>=8.0.0; platform_system != ""Windows""",Neutral,0.0
6,"pyreadline==2.1; platform_system == ""Windows""",Neutral,0.0
7,hikerapi==1.7.1,Neutral,0.0
1,requests-toolbelt==0.9.1,Neutral,0.0
2,geopy>=2.0.0,Neutral,0.0
3,prettytable==0.7.2,Neutral,0.0
4,instagram-private-api==1.6.0,Neutral,0.0
5,gnureadline>=8.0.0,Neutral,0.0
6,hikerapi==1.7.1,Neutral,0.0
1,I love this photo!,Positive,0.625
2,This post is boring.,Negative,-1.0
3,"It's okay, not great.",Neutral,0.05
4,Such a beautiful picture.,Positive,0.425
5,I'm not impressed.,Negative,-0.5
(venv) yuktha@yukthasri:~/Projects/Osintgram$
```

Result and Discussions:

5.1 Overview of the Results

The project aimed to perform **information security analysis** and **sentiment evaluation** on user-generated Instagram comments.
Using Python-based sentiment analysis, the system successfully processed the input text files and generated:

- A complete **sentiment classification table** (sentiment_report.csv)
- **Polarity scores** for each user comment
- **Graphical visualizations** (bar chart and pie chart)
- Categorization into **Positive, Negative, and Neutral** sentiments

This section evaluates the quality of the system's output and the insights derived from the data.

5.2 Sentiment Classification Results

The sentiment analysis produced a total of **18 processed comments**.
The sentiments were distributed as follows:

Sentiment	Count
Neutral	14
Positive	2
Negative	2

This result was confirmed by both:

- The terminal output
- The sentiment_report.csv file

- The generated visualizations (sentiment_bar_chart.png, sentiment_pie_chart.png)

5.3 Interpretation of Results

1. Neutral Sentiment Dominance

A major portion (about **78%**) of comments were categorized as *Neutral*.
This typically indicates:

- Informational or simple textual inputs
- Lack of emotional tone
- Comments that neither express praise nor criticism

Neutral dominance is expected in general social media datasets and shows that the model is correctly identifying comments with minimal polarity.

2. Positive Sentiment (≈11%)

Two comments displayed **positive polarity**, indicating:

- Appreciation
- Agreement
- Encouraging or favorable tone

This suggests that the system effectively detects text expressing positive emotion.

3. Negative Sentiment (≈11%)

Two comments were categorized as **negative**, which generally include:

- Criticism
- Dislike
- Disagreement
- Emotionally charged negativity

This shows the system is responding accurately to clearly negative wording.

5.4 System Performance Discussion

Accuracy and Reliability

TextBlob, being a rule-based and lexicon-based sentiment analyzer, provides fairly accurate outputs for:

- Simple sentences
- Basic opinionated text
- Short comments

In the dataset used, TextBlob correctly identified polarity variations, proving it reliable for lightweight sentiment tasks.

Conclusion and Future Development:

Conclusion:
This project demonstrates the practical application of Open-Source Intelligence (OSINT) tools in conducting forensic analysis on Instagram. Through the systematic collection and examination of publicly available data such as posts, comments, and user information, it establishes an effective framework for detecting suspicious activities and gathering digital evidence. The approach ensures that the investigation remains ethical, accurate, and in line with platform policies.

By integrating Python-based automation and analytical techniques, the system enhances the efficiency and reliability of digital investigations. The findings emphasize the growing significance of social media forensics in modern cybersecurity and law enforcement. Overall, the project illustrates how open-source tools can be effectively utilized to support secure, transparent, and legally compliant evidence collection and analysis in the digital environment.

Future Development:
In the future, this project can be enhanced by incorporating Machine Learning and Artificial Intelligence techniques to automatically detect unusual activities and hidden behavioral patterns within social media data. Implementing a centralized, real-time dashboard for

visualizing forensic results would improve usability and investigative efficiency. The framework could also be expanded to analyze data across multiple social media platforms, increasing its coverage and accuracy. Additionally, integrating cloud-based storage and continuous monitoring would make the system more scalable and adaptable for real-world digital forensic investigations. These enhancements would strengthen the system's reliability, performance, and overall effectiveness in modern cyber and forensic applications.

References:

1) https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2025.1566513/full
2) https://www.inderscienceonline.com/doi/10.1504/IJESDF.2025.147160
3) https://www.sciencedirect.com/science/article/pii/S1877050925015868
4) https://slogix.in/digital-forensics/latest-research-papers-in-digital-forensics/
5) https://www.heraldopenaccess.us/openaccess/digital-forensics-confronting-modern-cyber-crimes-technological-advancements-and-future-challenges
6) https://www.inderscience.com/info/inarticle.php?artid=147160
7) https://www.sciencedirect.com/science/article/pii/S1110866524000653
8) https://www.forensicscijournal.com/articles/jfsr-aid1074.php
9) https://www.nbftools.com/forensic-investigations-of-social-media-apps-tools-and-techniques/
10) https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4941525